

PREDICTING LENGTH OF STAY IN A NEONATAL INTENSIVE CARE UNIT

By

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Abstract

Neonatal intensive care units (NICUs) provide specialized treatment for infants born prematurely or with serious medical complications, and accurate estimation of length of stay is essential for effective clinical planning and resource management. This study examines the use of machine learning models to predict NICU length of stay and applies k-means clustering to identify clinically meaningful patient subgroups. The data consist of de-identified patient records from Duke University Hospital and include gestational age, birth weight, demographic characteristics, and major neonatal morbidities. Since length of stay has a strongly right-skewed distribution, a log-linear regression model serves as a baseline and its performance in terms of root mean squared error is compared to that of a random forest model. The random forest approach yields more accurate estimates of hospitalization time, particularly after limiting the influence of extreme values. In addition, k-means clustering provides further insight into the factors that affect length of stay by grouping infants according to shared clinical characteristics. The results reveal distinct populations that differ substantially in gestational age, morbidity status, and average length of stay.

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Introduction

Neonatal intensive care units (NICUs) play a critical role in caring for the most vulnerable newborns, those born prematurely or with severe medical conditions. Clinical outcomes in a NICU depend on a complex interaction of biological and environmental factors such as gestational age, birth weight, and the presence of neonatal complications. NICU care is both resource-intensive and vital to infant survival, which is why hospitals and researchers have long sought to understand which factors most strongly influence length of stay and clinical outcomes [5].

The goal of this analysis is to explore how gestational age, birth weight, and significant neonatal morbidities relate to both length of stay and survival outcomes in the NICU setting. By identifying patterns in this data, we can gain insight into how specific clinical factors affect hospital efficiency, patient prognosis, and overall quality of care.

I find this dataset particularly meaningful because it highlights how data science and medicine can intersect to improve patient care. Behind each data point is a fragile newborn whose outcome reflects not only medical treatment but also systemic factors like resource allocation and hospital policy. Using data to identify where improvements can be made is a powerful reminder that research in healthcare analytics is ultimately about supporting life in its earliest and most vulnerable form.

Dataset Description

The dataset used in this project consists of de-identified NICU patient data from Duke University Hospital. All identifying information was removed to protect patient privacy. The data were originally collected to support the development of a neonatal intensive care unit simulation model created by consultants at the SAS Institute in collaboration with physicians at Duke University Hospital [4, 6]. The primary purpose of this simulation model was to provide a safe and ethical framework for examining relationships among length of stay, resource utilization, and clinical outcomes in the NICU setting [4].

The primary response variable examined in this analysis is length of stay, measured as the total number of days an infant remained hospitalized in the NICU. Longer NICU stays are often associated with prematurity, lower birth weight, and the presence of serious neonatal complications. Many severe neonatal conditions require increased levels of clinical monitoring and nursing care, including lower nurse-to-patient ratios. As a result, examining length of stay across different morbidities provides insight into both clinical burden and hospital resource utilization.

Several clinical and demographic predictors are included in the dataset. Gestational age, measured in weeks at birth, reflects the degree of infant maturity and is one of the strongest predictors of neonatal outcomes. Birth weight, recorded in grams, provides additional information about infant development and vulnerability. Demographic variables include infant sex, race or ethnicity, and inborn status, which indicates whether the infant was born at Duke University Hospital (“inborn”) or transferred from another facility (“outborn”).

The dataset also includes indicators for several major neonatal morbidities known to substantially affect NICU outcomes and length of hospitalization. These conditions include

intraventricular hemorrhage (IVH), a form of bleeding within the brain that varies in severity by grade; retinopathy of prematurity (ROP), a disorder involving abnormal blood vessel development in the eye; necrotizing enterocolitis (NEC), a severe gastrointestinal disease that may require medical or surgical treatment; and patent ductus arteriosus (PDA), a cardiovascular condition that can be managed medically or surgically. Each of these morbidities has been shown in prior research to contribute to increased clinical complexity and extended NICU length of stay [4, 5].

Literature Review

Predictive Modeling and Machine Learning

Predictive modeling, or predictive analytics, is about using data and statistical algorithms to predict what might happen next, given the current process and environment [8]. There are two types of predictive models: classification and regression. In simple terms, classification models predict discrete categories, whereas regression models predict continuous values [7].

Machine learning falls under the umbrella of predictive analysis and is based on the idea that systems can learn from data, identify patterns, and make decisions with minimal human intervention [10]. It can be applied in various contexts; however, it is commonly categorized into two primary categories: supervised and unsupervised learning. With supervised methods, you are interested in predicting values of an output variable based on a collection of input variables [8]. The term “supervised” indicates that the response variable guides the learning process [2]. In contrast, unsupervised learning includes a set of statistical tools to better understand and describe your data, but performs the analysis without a target variable. This approach is commonly used in exploratory data analysis rather than for prediction. Thus, while the primary focus of this project is supervised learning to predict a response variable, unsupervised methods are also incorporated to explore underlying patterns and groupings within the data.

Predictive modeling through machine learning begins by splitting the dataset into a training set and a testing set. The training set is used to fit models and compare different modeling approaches before selecting a final model. The test set remains untouched during model selection and is used only after the final model has been selected. The test set is then used to obtain an unbiased assessment of the model’s performance, also referred to as the generalization error [2]. For this analysis, the full dataset was split using simple random

sampling with an 80–20 split. This means that 80% of the observations were randomly assigned to the training set, while the remaining 20% were assigned to the testing set.

In addition to splitting the dataset, k -fold cross-validation is used during model training. This resampling method provides an alternative approach by repeatedly fitting models to subsets of the training data and evaluating them on other subsets. The training data are randomly divided into k groups (folds) of approximately equal size [2]. The model is trained on $k - 1$ folds and validated on the remaining fold, and this process is repeated until each fold has served as the validation set once. For this dataset, 10-fold cross-validation is used, as it provides a reliable balance between computational efficiency and stable performance estimation for large datasets. In this analysis, it is specifically applied during the training of the random forest model to assist in selecting optimal tuning parameters, such as the number of variables considered at each split and the number of trees. By evaluating different parameter combinations across multiple folds, this approach reduces the risk of overfitting and results in a more reliable estimate of model performance.

Linear Regression

Linear regression is one of the most widely used supervised learning methods and serves as a foundational technique in predictive modeling [2]. Regression models are used when the response variable is continuous, which aligns with the goal of this project to predict length of stay. Linear regression models the relationship between one or more predictor variables and a response variable by assuming this relationship can be approximated by a linear function. In its simplest form, a linear regression model describes the relationship between a single predictor variable X and a response variable Y . The model is written as

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

where Y represents the observed response value, X represents the predictor value, β_0 is the intercept, β_1 is the slope coefficient, and ϵ represents random error [2]. The intercept represents the expected value of the response when the predictor is zero, while the slope represents the expected change in the response for a one-unit increase in the predictor variable.

Many real-world problems involve more than one predictor variable. Multiple linear regression extends the simple linear regression framework to allow for two or more predictors. The general multiple linear regression model with p predictors is expressed as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon.$$

This framework allows the model to account for the combined effects of multiple clinical and demographic variables when predicting NICU length of stay.

The coefficients in a linear regression model are most commonly estimated using the method of least squares. This approach selects coefficient values that minimize the residual sum of squares (RSS), which measures the total squared difference between observed response values and the values predicted by the model. The residual sum of squares is defined as

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

where y_i is the observed response and $\hat{y}_i$ is the corresponding predicted value [2]. Smaller RSS values indicate that the model's predictions are closer to the observed outcomes, resulting in a better overall model fit.

One advantage of linear regression is its interpretability. Each coefficient provides a clear, quantitative measure of how a predictor is associated with changes in the response variable. Linear regression is also a parametric method, meaning it relies on assumptions about the functional form of the relationship between predictors and the response [7]. While this structure

allows for straightforward interpretation, it also limits flexibility, as linear regression may struggle to capture complex or nonlinear relationships present in the data. Despite these limitations, linear regression serves as an important baseline model in this analysis.

Decision Trees and Random Forests

Decision trees are another type of supervised learning method used for both classification and regression problems. In regression problems, decision trees predict a continuous response by partitioning the predictor space into smaller, non-overlapping regions using a sequence of decision rules [2]. These rules form a hierarchical tree structure composed of nodes and branches. The process begins at the root node, which contains all training observations, and proceeds through internal nodes where the data are repeatedly split based on predictor values. The final subgroups at the bottom of the tree are referred to as terminal nodes, or leaves, and each leaf represents a region of the predictor space with a single predicted response value. The connections between nodes are called branches and represent the decision paths taken through the tree [2].

Decision trees are considered nonparametric models because they do not assume a specific functional form between the predictors and the response variable. This allows them to capture nonlinear relationships and interactions between variables without requiring these relationships to be specified in advance [2]. Another advantage of decision trees is their interpretability. The hierarchical structure of the tree makes it possible to visually and conceptually follow how predictions are made, which can be useful when understanding the influence of individual predictors [12].

Despite these advantages, decision trees often have limited predictive accuracy when used on their own. Trees are sensitive to the training data, and small changes in the data can lead

to different tree structures, which increases variance and can reduce performance on unseen data [2]. This limitation motivates the use of ensemble methods, such as random forests.

Random forests are an extension of decision trees that improve predictive performance by combining many individual trees into a single model. Each tree in a random forest is trained on a bootstrapped sample of the training data, meaning that observations are sampled with replacement to create multiple datasets [7]. In addition, at each split within a tree, only a randomly selected subset of predictors is considered. This additional randomness reduces correlation among the trees and increases model stability.

For regression problems, the final prediction from a random forest is obtained by averaging the predictions across all trees in the ensemble. By aggregating many decorrelated trees, random forests reduce overfitting and generally achieve higher predictive accuracy than a single decision tree [7]. Although random forests are less interpretable than individual trees, they still allow for analysis of variable importance, which can help identify predictors that have a strong influence on the response variable.

Data Wrangling

Before beginning the exploratory data analysis and modeling stages of this project, several data wrangling steps needed to be done. Data wrangling involves cleaning and organizing raw data so that variables are consistent, readable, and appropriate for statistical methods [1]. In this project, wrangling included removing unnecessary variables, renaming columns, recoding categorical values, and examining missing data.

The first step was removing variables that were either redundant or not relevant to the goals of this analysis. The original dataset contained two birth weight columns (`birth_weight` and `BWgt`) with identical values, so one was removed (`BWgt`). Similarly, both full-text and abbreviated versions of the retinopathy of prematurity (ROP) and patent ductus arteriosus (PDA) variables were included. The abbreviated versions were kept because they contained the same information in a more consistent format. The variable `LOS_Predictor` was also removed, as it was created by the original dataset author (Dr. Chris DeRienzo) for his own exploratory purposes.

Next, column names were cleaned and standardized to a consistent `snake_case` (all lowercase letters separated by underscores) format to improve readability. Several variables were also renamed so their meanings were clearer and easier to work with during the exploratory analysis stage. For example, categorical variables were recoded from abbreviated or all-capital values into descriptive labels to make results and visualizations easier to interpret.

Missing values were then examined across all variables. The numeric predictors, `gestational_age` and `birth_weight`, contained no missing values. However, the clinical condition variables `ivh`, `nec`, and `pda` did include missing observations. Rather than removing these rows, they were relabeled as “Missing” so that all observations could be explored during

the exploratory analysis stage. Summary statistics were also generated for the numeric variables to confirm that values fell within clinically reasonable ranges. Finally, all categorical variables were converted to factors to ensure proper handling during the exploratory analysis and modeling stages.

Exploratory Data Analysis (EDA)

After wrangling and cleaning the data, an exploratory analysis was conducted using summary statistics and visualizations to better understand the variables included in this dataset. As an initial step, the response variable was examined independently to assess its distribution and suitability for modeling. In this analysis, the variable `length_of_stay` was selected as the response variable.

Statistic	Value (Days)
n	4,297
Mean	25.70
Standard Deviation	40.76
Median	9
IQR	22

Table 1. Summary statistics for length of stay (in days).

As shown in Figure 1, the distribution of length of stay is unimodal and strongly right-skewed. The accompanying boxplot highlights the presence of multiple outliers in the upper range, further reinforcing the extent of the right skew. This indicates that while many infants experience relatively short NICU stays, a smaller subset remains hospitalized for substantially longer periods. Summary statistics for length of stay are provided in Table 1. The median length of stay is 9 days, and the interquartile range (IQR) of 22 days indicates that the middle 50% of observations fall between 5 days (Q1) and 27 days (Q3). In contrast, the mean length of stay is 25.70 days, which is substantially higher than the median due to the influence of a small number of infants with very long hospitalizations. This difference between the mean and median further reflects the strong right skew in the distribution.

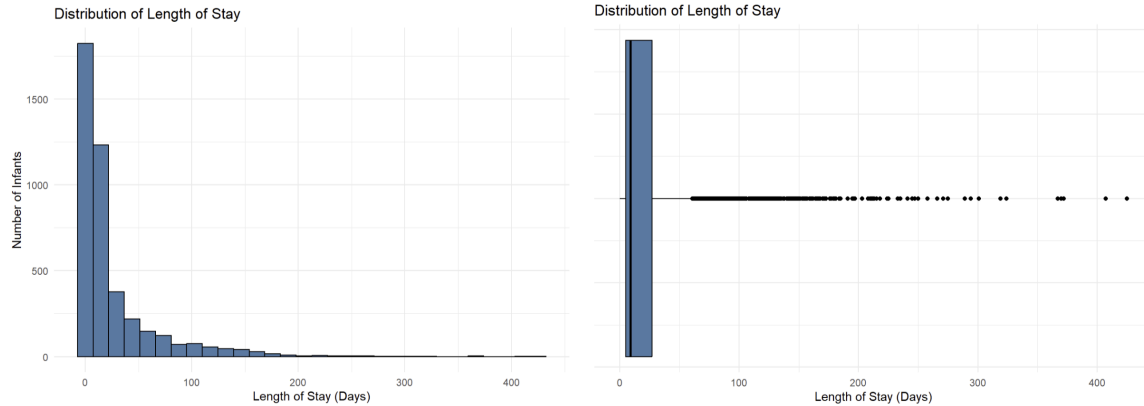


Figure 1. The response variable (`length_of_stay`) is heavily right-skewed.

Due to the pronounced right skew observed in the response variable, a natural log transformation was applied to length of stay to improve symmetry and reduce the influence of extreme values. Figure 2 displays the histogram and boxplot of the log-transformed length of stay. After transformation, the distribution appears more symmetric, making it more appropriate for use in the modeling stage.

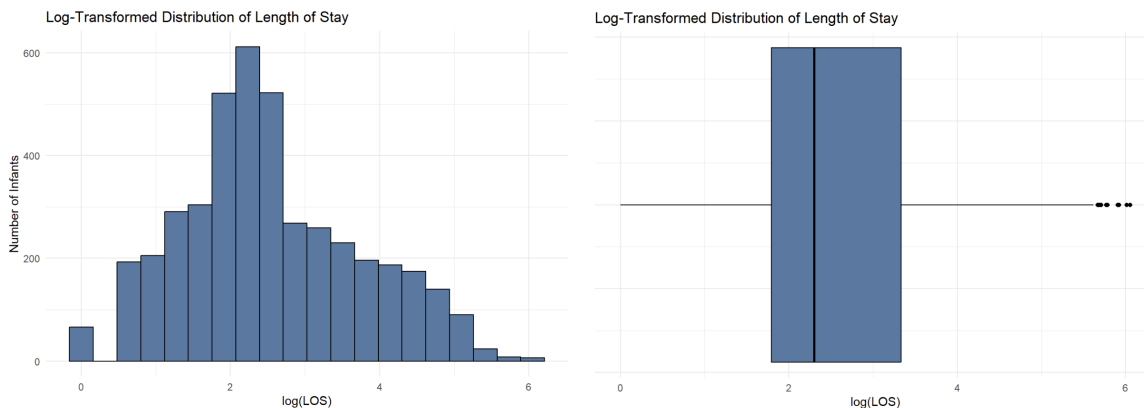


Figure 2. The log-transformed distribution of length of stay is closer to symmetric.

Next, the relationship between the response variable (`length_of_stay`) and other clinical variables was explored. Examining these relationships during the exploratory data analysis stage helps identify patterns that are later assessed more formally through predictive models. One of the most influential clinical variables examined was gestational age, measured in weeks at birth. A full-term pregnancy typically ranges from 38 to 42 weeks, with infants born

before 37 weeks considered premature and those born after 42 weeks considered postmature [12]. As shown in Figure 3, the scatter plot displays the relationship between gestational age and length of stay for individual infants. A clear negative association is observed, with lower gestational ages generally corresponding to longer NICU stays. However, extreme outliers are present across a range of gestational ages, indicating that prolonged hospitalizations are not limited to the most premature infants.

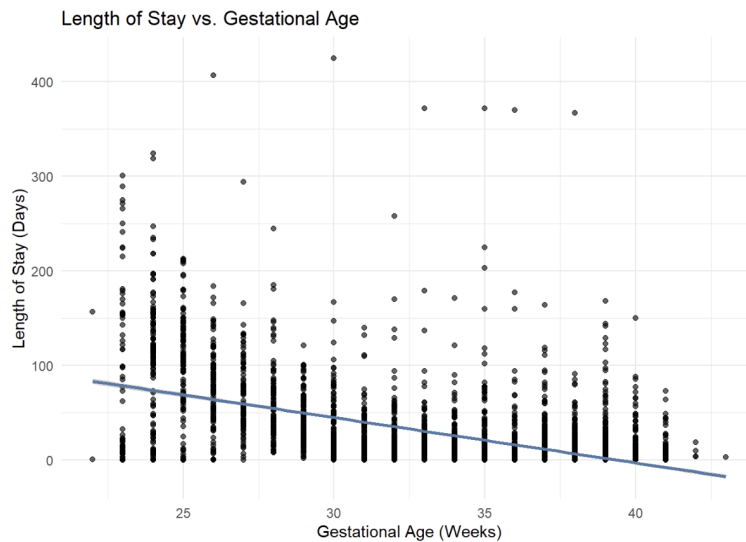


Figure 3. Length of stay decreases with increasing gestational age, with greater variability at lower gestational ages.

As shown in Figure 4, the line plot of average length of stay provides a clearer view of the overall trend between gestational age and NICU hospitalization time. By summarizing the data across gestational age, the variability observed in Figure 3 is reduced, making the underlying pattern more apparent. The average length of stay is highest among infants born at the earliest gestational ages, with a peak observed around 24 weeks. As gestational age increases, the average length of stay declines steadily, reinforcing the strong inverse relationship between gestational age and length of stay. This pattern suggests that infants born closer to full term generally require shorter NICU stays, while more premature infants experience prolonged hospitalization due to increased medical complexity.



Figure 4. Average length of stay decreases as gestational age increases.

Figure 5 shows the distribution of gender in the dataset. The sample includes slightly more male infants than female infants, although both groups are well represented. This supports the inclusion of gender as a predictor in the modeling stage, as there are sufficient observations in each group.

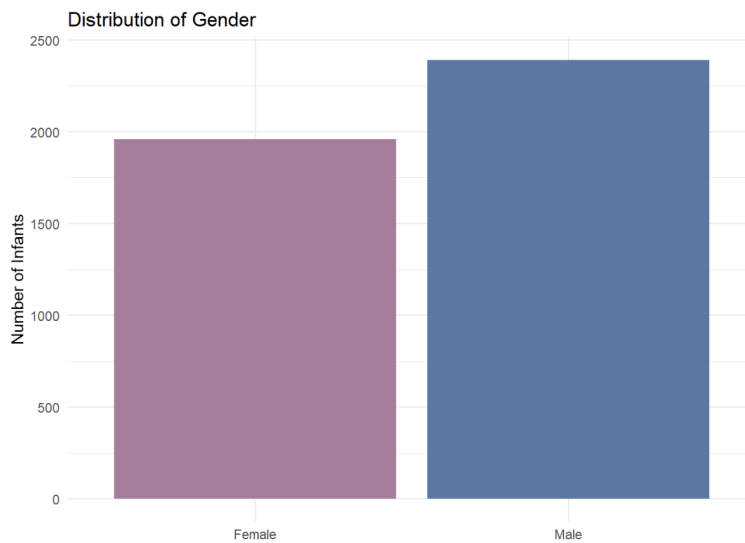


Figure 5. There are slightly more male infants than female infants.

Figure 6 shows the distribution of race and ethnicity in the dataset. The majority of infants are classified as White, followed by Black infants, with a smaller number of infants

categorized as Other. While all groups are represented, the distribution is somewhat imbalanced, which may influence the interpretation of race and ethnicity in the modeling stage.

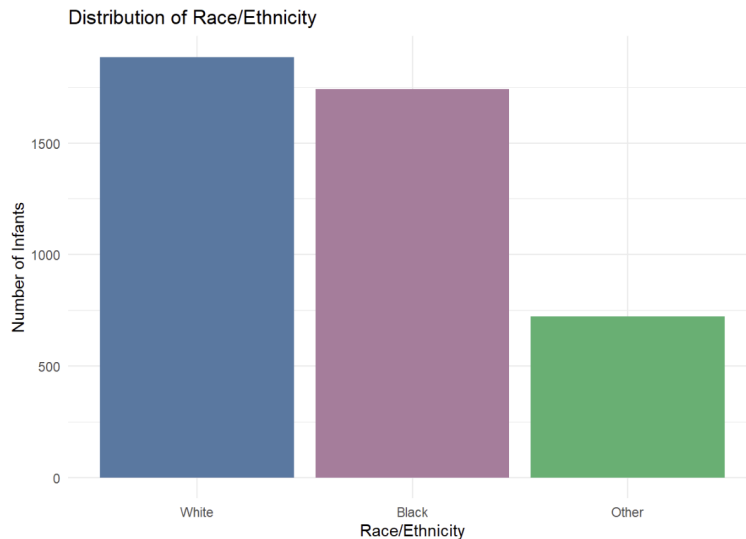


Figure 6. White and Black infants make up the majority of the sample, with fewer infants in the Other category.

The next set of figures further explores the relationship between the response variable, `length_of_stay`, and major neonatal morbidities. Figure 7 focuses on intraventricular hemorrhage (IVH), a condition characterized by bleeding into the internal structures of the brain. As shown in Figure 7, the typical length of stay for infants diagnosed with IVH (as measured by the median value) is longer compared to infants without IVH. However, differences in variability are also evident across categories. Infants with Grade I and Grade II IVH exhibit larger overall ranges of length of stay, indicating greater variability in hospitalization time, while infants with Grade II and Grade IV IVH have the largest interquartile ranges, indicating that the middle 50% of observations span a wider range of hospitalization times than the other grades. In contrast, infants without IVH have a smaller IQR, although the presence of numerous outliers suggests that prolonged hospitalizations can still occur in the absence of IVH. Among the IVH categories, the highest median length of stay is observed for infants with Grade II IVH, followed by those

with Grade IV IVH. All distributions appear right-skewed, reflecting the influence of extreme values. Although IVH is associated with longer and more variable lengths of stay, the frequency table (Table 2) indicates that relatively few infants fall into the higher severity categories, with only 4.64% classified as Grade I and 1.52% as Grade IV. Grade IV IVH represents the most severe form of the condition, involving bleeding that extends into surrounding brain tissue and typically requiring more intensive monitoring and prolonged medical care.

IVH Category	n	%
None	3,964	91.13
Grade I	202	4.64
Grade II	48	1.10
Grade III	22	0.51
Grade IV	66	1.52

Table 2. Frequency distribution of intraventricular hemorrhage (IVH).

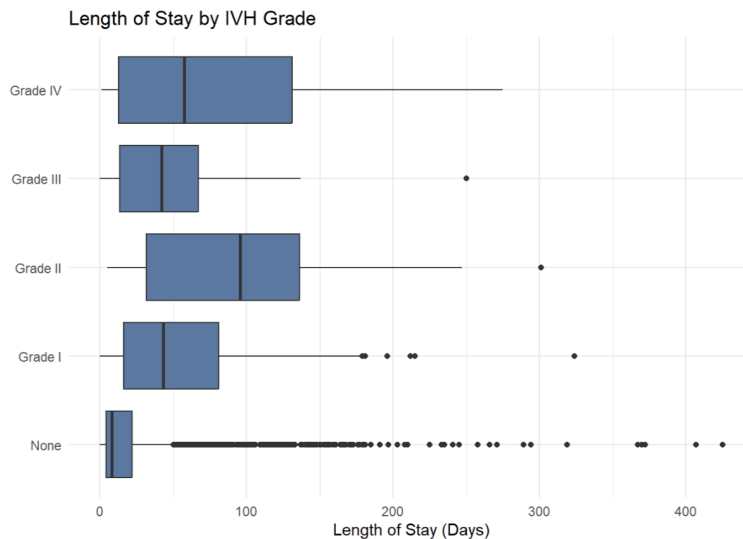


Figure 7. The longest median lengths of stay are observed among infants with Grade II and Grade IV IVH.

Figure 8 examines the relationship between length of stay and retinopathy of prematurity (ROP), a condition involving abnormal development of blood vessels in the eye that can lead to

vision impairment or blindness if not properly treated. ROP severity is categorized by stage, ranging from no ROP to advanced Stage III–IV disease, with or without surgical intervention. As shown in Figure 8, the median length of stay increases steadily as ROP severity progresses, indicating that infants with more severe ROP tend to experience longer NICU stays. Differences in variability are also observed across ROP stages. Infants with Stage II ROP exhibit the largest overall range of length of stay, suggesting greater variability in hospitalization time. In contrast, infants without ROP have a smaller IQR, although the presence of numerous outliers indicates that prolonged stays can still occur in the absence of this condition. Similarly to IVH, all ROP categories appear right-skewed, reflecting a subset of infants with particularly extended hospitalizations. Although higher stages of ROP are associated with longer and more variable lengths of stay, Table 3 shows that only a small percentage of infants in the dataset are affected by ROP, with about 90% classified as having no ROP.

ROP Category	n	%
None	3,943	90.64
Stage I	102	2.34
Stage II	189	4.34
Stage III-IV (No Cyro or Laser Surgery)	19	0.44
Stage III-IV (With Cyro or Laser Surgery)	97	2.23

Table 3. Frequency distribution of retinopathy of prematurity (ROP).

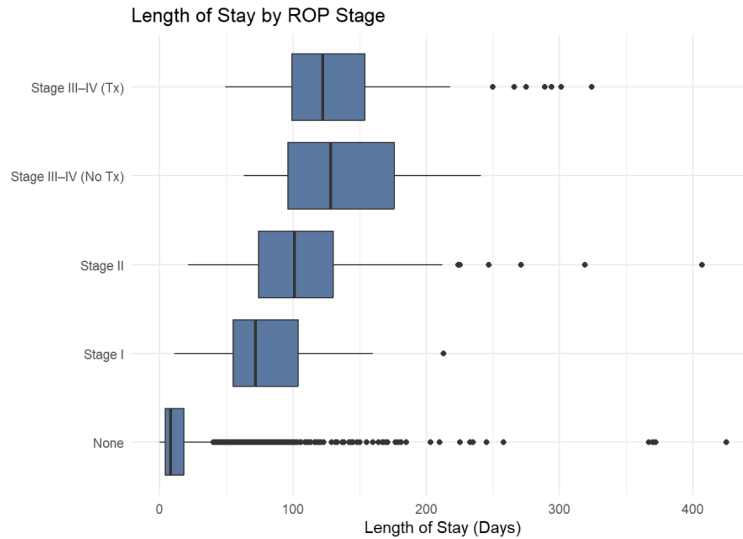


Figure 8. Median length of stay increases as ROP severity increases.

Figure 9 examines the relationship between length of stay and necrotizing enterocolitis (NEC), a severe neonatal intestinal condition categorized as none, medical, or surgical. As shown in Figure 9, infants diagnosed with NEC experience substantially longer median lengths of stay compared to those without the condition. Among affected infants, those treated medically have the highest median length of stay, while those who underwent surgical treatment have slightly shorter median stays. In contrast, infants without NEC have much lower median lengths of stay, indicating shorter and less complex hospitalizations. The distributions also differ in spread and shape across categories. The group without NEC exhibits the largest overall range of length of stay, reflecting the presence of several extreme values, while the medical and surgical NEC groups have wider interquartile ranges, indicating greater variability in hospitalization time among affected infants. Although the group without NEC has a smaller interquartile range, the presence of numerous outliers suggests that prolonged stays can still occur in the absence of this condition. All NEC categories appear right-skewed, reflecting the influence of these extreme values. Although NEC is associated with longer and more variable lengths of stay, Table 4 shows

that the vast majority of infants in the dataset do not experience NEC, with over 96% classified in the “None” category.

NEC Category	n	%
None	4,199	96.53
Medical	63	1.45
Surgical	39	0.90

Table 4. Frequency distribution of necrotizing enterocolitis (NEC).

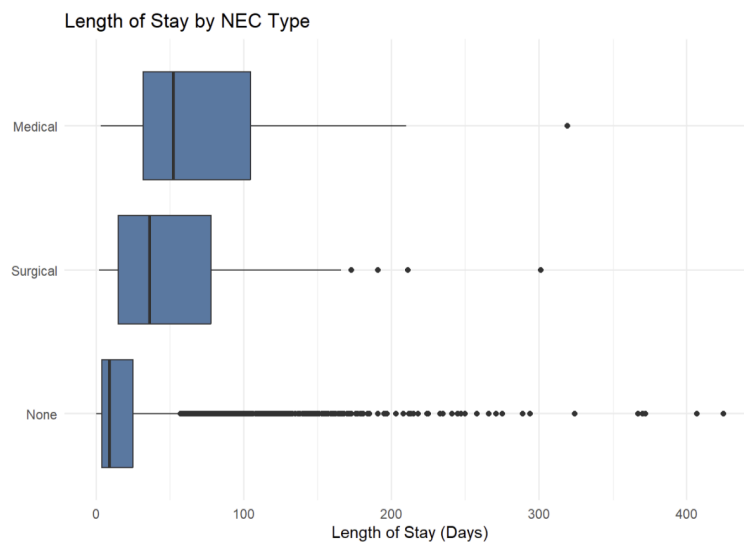


Figure 9. Infants with medically managed NEC have longer median lengths of stay than those with surgically treated NEC.

Lastly, Figure 10 examines the relationship between length of stay and the presence and management of patent ductus arteriosus (PDA), a neonatal cardiovascular condition that may require medical or surgical intervention. As shown in the figure, infants who underwent surgical ligation of PDA experienced the longest median length of stay. Infants managed with medication also have higher median lengths of stay compared to those without PDA, while infants without the condition have the lowest median length of stay overall, indicating shorter and less complex hospitalizations. Differences in variability are also evident across categories. Infants with PDA, particularly those requiring treatment, exhibit wider interquartile ranges and larger overall ranges

of length of stay, indicating greater variability in hospitalization time. In contrast, similar to the other morbidities, infants without PDA have a smaller interquartile range, although the presence of multiple outliers in the upper range suggests that prolonged stays can still occur in the absence of this condition. The distributions for the PDA categories also appear right-skewed, with the exception of the “None (Medication Only)” group. As shown in Table 5, the majority of infants in the dataset do not require treatment for PDA, with approximately 77% classified as having no PDA, while only small proportions receive medical treatment (4.71%) or surgical ligation (2.69%). However, a notable portion of infants (14.05%) have PDA that is not treated, suggesting that many cases may not require active intervention.

PDA Category	n	%
None	3,358	77.20
None (Medication Only)	12	0.28
Yes (Untreated)	611	14.05
Yes (Medically Treated)	205	4.71
Yes (Ligated)	117	2.69

Table 5. Frequency distribution of patent ductus arteriosus (PDA).

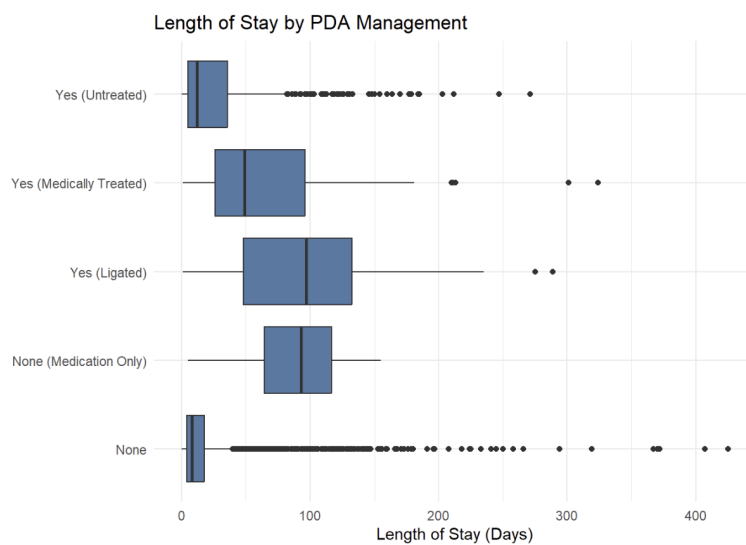


Figure 10. Length of stay differs across PDA management categories.

K-Means Clustering

In addition to predictive modeling techniques, an unsupervised learning method was applied to explore whether natural groupings of infants existed within the dataset. Clustering is a statistical technique used to identify patterns within data by grouping observations that share similar characteristics. Unlike supervised learning approaches, clustering does not rely on a response variable and instead focuses on uncovering patterns that naturally emerge within the predictors themselves [3]. These clusters represent collections of observations that are more similar to one another than to observations in other groups, capturing underlying relationships that may not be immediately apparent within the dataset [3].

One of the most widely used clustering methods is k-means clustering, which partitions observations into a predetermined number of groups, denoted by k . The goal of the method is to divide the observations into mutually exclusive clusters such that observations within the same cluster are as similar as possible while observations in different clusters are as dissimilar as possible [2]. Each cluster is represented by a central point known as a centroid, which corresponds to the mean value of all observations assigned to that cluster [2].

The k-means algorithm operates through an iterative process. First, the number of clusters k is specified by the analyst. The algorithm then randomly selects k observations from the dataset to serve as initial cluster centers. Each observation is assigned to the nearest centroid based on a distance measure, most commonly Euclidean distance. After the assignment step, the centroid of each cluster is recalculated using the mean values of the observations assigned to that cluster. The algorithm alternates between assigning observations to clusters and updating cluster centroids until the cluster assignments stabilize and the total within-cluster variation is minimized [2]. The objective of the algorithm is to construct clusters that minimize the total

within-cluster variation, which measures how far observations within each cluster are from their corresponding cluster centroid [2]. Smaller values indicate more compact clusters and greater similarity among observations within each group.

K-means clustering relies on distance calculations, so the variables used in the clustering procedure must be numeric and measured on comparable scales. Variables measured in different units can disproportionately influence the distance calculations and bias the clustering results. To address this issue, all predictor variables were standardized using the z-score transformation prior to performing the clustering analysis. Standardization ensures that each variable contributes equally to the distance calculations regardless of its original scale. The clustering analysis focused on six clinically relevant variables: gestational age, birth weight, and the presence or absence of four major neonatal morbidities (intraventricular hemorrhage, retinopathy of prematurity, necrotizing enterocolitis, and patent ductus arteriosus). The morbidity indicators were categorical variables, so they were converted to binary numeric values before applying the clustering algorithm so that they could be included in the distance calculations.

Determining an appropriate number of clusters is an important step when applying k-means clustering. In practice, the number of clusters is often chosen by the analyst based on exploratory methods and interpretability [2]. One commonly used technique is the elbow method, which evaluates how the total within-cluster sum of squares changes as the number of clusters increases. As additional clusters are added, the within-cluster variation decreases; however, after a certain point the improvement becomes marginal. The point where this reduction begins to level off is referred to as the “elbow” and provides a reasonable estimate for the number of clusters to use [2].

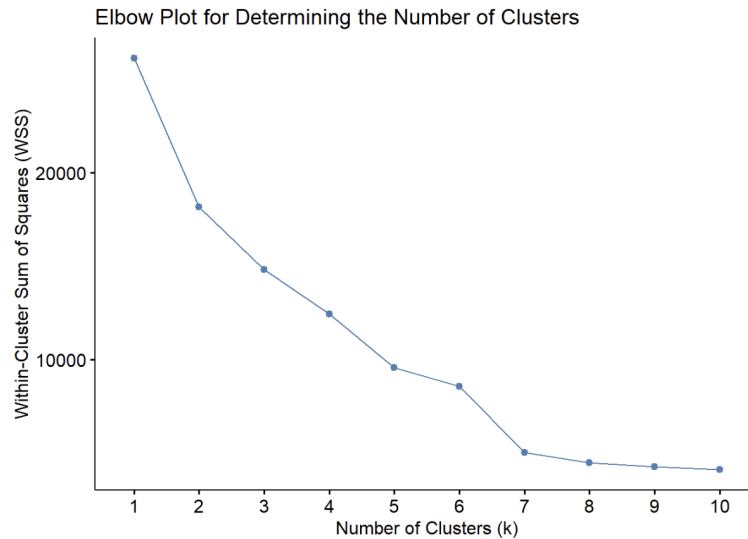


Figure 11. Within-cluster variation decreases as the number of clusters (k) increases, leveling off around 7.

Figure 11 displays the elbow plot used to evaluate the appropriate number of clusters for the NICU dataset. The plot shows the within-cluster sum of squares for increasing values of k . Although the curve begins to level off around $k = 7$, several clustering solutions were evaluated for $k = 2, 4, 5$, and 7 to assess interpretability. Among these options, the four-cluster solution produced clusters that were both clearly separated and clinically meaningful.

To facilitate interpretation, the clusters were reordered according to mean gestational age and labeled to reflect increasing levels of prematurity. The resulting groups included extremely premature infants, very premature infants, moderately premature infants, and near-term infants.

Cluster	Mean Gestational Age (Weeks)	Mean Birth Weight (Grams)	IVH Rate	ROP Rate	NEC Rate	PDA Rate	Mean Length of Stay (Days)
Extremely Premature	26.92	994	50.83%	69.17%	0%	42.39%	84.72
Very Premature	28.76	1,312	25.49%	29.41%	100%	42.16%	69.90
Moderately Premature	32.31	1,786	0.11%	0%	0%	13.67%	19.86
Near-Term	38.19	3,304	1.82%	0%	0%	22.09%	11.50

Table 6. Clinical characteristics of infants across the four clusters identified by the k-means clustering analysis.

Table 6 summarizes the average clinical characteristics for each cluster, including gestational age, birth weight, neonatal morbidity rates, and length of stay. These summary statistics provide an overview of how the clusters differ across key clinical variables.

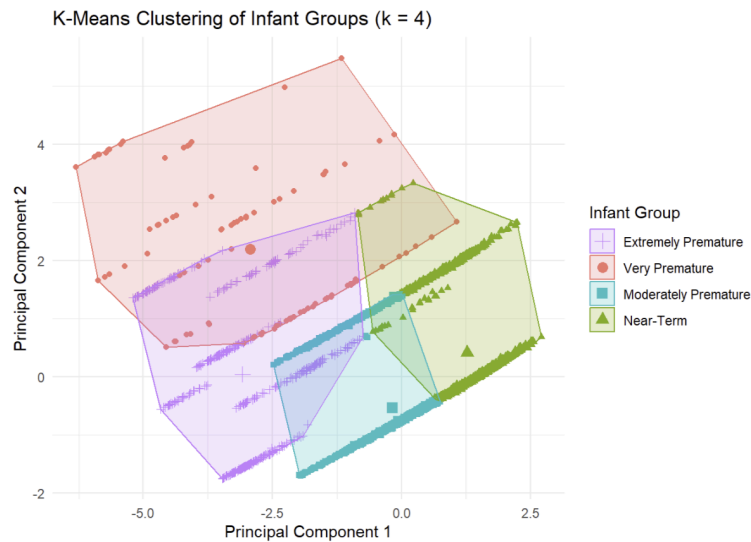


Figure 12. K-means clustering of infants based on gestational age, birth weight, and major clinical complications, identifying four prematurity groups.

Figure 12 presents a visualization of the four-cluster k-means solution. Each point represents an individual observation projected onto the first two principal components, allowing the clusters to be visualized in two-dimensional space. The figure demonstrates that the

observations form relatively distinct groupings, supporting the selection of four clusters for the final analysis.

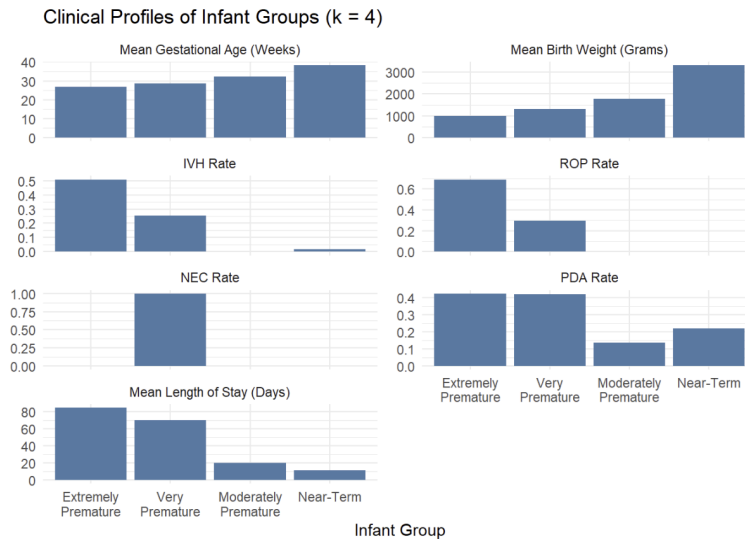


Figure 13. Infants with lower mean gestational age and birth weight have higher complication rates and longer NICU stays.

To further illustrate how the clusters differ clinically, Figure 13 displays the cluster profile plot. This plot compares the average values of the selected variables across the four clusters, highlighting differences in gestational age, birth weight, neonatal morbidity rates, and length of stay.

The clustering results reveal that infants in the extremely premature group exhibited the lowest gestational ages and birth weights, along with the highest rates of neonatal complications and the longest hospital stays. In contrast, the near-term group consisted of infants with substantially higher gestational ages, minimal complication rates, and much shorter NICU stays. Notably, no infants in the moderately premature or near-term groups exhibited retinopathy of prematurity (ROP) or necrotizing enterocolitis (NEC), further distinguishing these groups from the more premature clusters. The intermediate clusters captured varying levels of prematurity and clinical complexity. Overall, the clustering analysis suggests that gestational age and

associated neonatal complications naturally partition the NICU population into clinically meaningful subgroups.

Building the Models

Before building the predictive models, the dataset was divided into training and testing sets to allow for evaluation on unseen data. A multiple linear regression model was initially considered; however, because length of stay was strongly right-skewed, a natural log transformation was applied to the response variable. This resulted in a log-linear regression model using predictors including gestational age, birth weight, survival status, gender, inborn status, race/ethnicity, and indicators for major neonatal morbidities (IVH, ROP, NEC, and PDA).

m_{try}	Number of Trees	RMSE
4	300	26.67131
4	500	26.59733
4	800	26.69672
5	300	26.89861
5	500	26.94190
5	800	26.92929
6	300	27.11096
6	500	27.10498
6	800	27.14325

Table 7. Grid search results for random forest hyperparameter tuning.

To account for potential nonlinear relationships and interactions among predictors, a random forest model was also developed using the same set of variables. The random forest constructs multiple decision trees using bootstrapped samples and aggregates their predictions to improve accuracy. Two hyperparameters were explored for the random forest model: the number of variables considered at each split (m_{try}) and the number of trees in the model. The parameter m_{try} controls how many predictors are randomly selected at each split, helping to balance tree correlation and predictive strength [2]. The number of trees determines how many decision trees are included in the ensemble, where a larger number of trees helps stabilize error and improve

predictive accuracy [2]. These hyperparameters are known to have the greatest impact on model performance and are therefore important to tune [2]. A grid search was conducted across several combinations, and the best-performing model used $m_{try} = 4$ and 500 trees based on out-of-bag RMSE. The grid search was conducted using the full training dataset. The selected hyperparameters were then applied to both the full and reduced (90th percentile) datasets for model evaluation.

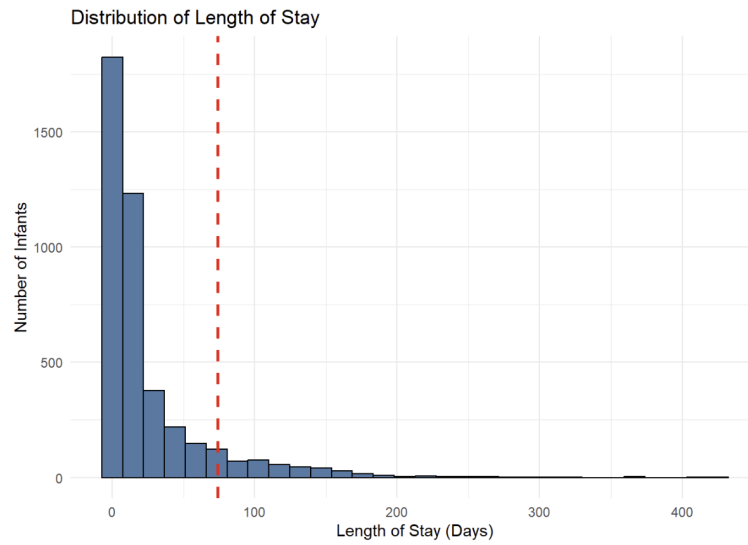


Figure 14. The 90th percentile cutoff (73 days) indicated by a dashed vertical red line.

There were a small number of infants with unusually long lengths of stay, which had a strong influence on model performance. To address this, an additional approach was considered that restricted the dataset to observations at or below the 90th percentile (73 days). Both the log-linear regression and random forest models were then applied to this restricted dataset to evaluate how model performance changed when extreme values were reduced.

Model Validation

Model accuracy was evaluated to determine how well the models predict length of stay for unseen data. To assess model performance, the root mean squared error (RMSE) was used. RMSE measures the average difference between predicted and observed values in the testing dataset. For the log-linear model, predictions were back-transformed to the original scale so that RMSE could be interpreted in days [2]. Lower RMSE values indicate better predictive accuracy, so the goal is to minimize this value. The RMSE can be calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2},$$

where y_i represents the observed values, $\hat{y}_i$ represents the predicted values, and n is the number of observations in the testing dataset.

The log-linear regression model applied to the full dataset resulted in an RMSE of 31.0 days, meaning that predictions were, on average, 31.0 days from the observed values. When applied to the dataset restricted to the 90th percentile, the RMSE decreased to 12.5 days. The random forest model performed better than the regression model in both cases. On the full dataset, the random forest achieved an RMSE of 29.4 days. When restricted to the 90th percentile, the random forest achieved the lowest RMSE of 11.1 days.

Model	RMSE (Days)
Log-Linear Regression (Full Data)	31.0
Log-Linear Regression (90th Percentile)	12.5
Random Forest (Full Data)	29.4
Random Forest (90th Percentile)	11.1

Table 8. RMSE values for each predictive model.

These results show that the random forest model produced lower prediction error than the log-linear regression model, and that model performance improved when the dataset was restricted to the 90th percentile of length of stay. To further interpret the random forest model, a variable importance plot was generated (Figure 15). This plot ranks predictors based on their contribution to reducing prediction error across the ensemble of trees. As shown in Figure 15, retinopathy of prematurity (ROP) is the most important predictor of length of stay, followed by birth weight and gestational age. These results highlight the strong influence of both clinical complications and developmental maturity on NICU hospitalization duration. In contrast, variables such as necrotizing enterocolitis (NEC), gender, and inborn status contribute relatively less to predictive performance. Overall, the variable importance findings reinforce earlier results, indicating that prematurity and associated morbidities are the primary drivers of length of stay.

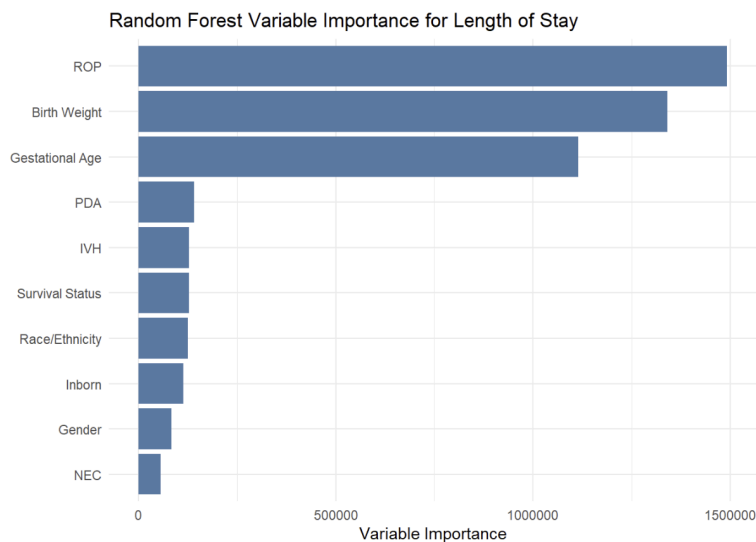


Figure 15. ROP, birth weight, and gestational age are the most important predictors of length of stay.

Conclusions

The random forest model proved to be a more accurate method for predicting NICU length of stay than the log-linear regression model, as measured by RMSE. This result was expected given the ability of random forests to capture nonlinear relationships and interactions among clinical variables. Restricting the dataset to the 90th percentile led to a substantial improvement in predictive accuracy for both models. This indicates that a small number of extreme observations have a strong influence on model performance. The presence of these unusually long hospital stays introduces variability that makes accurate prediction more difficult, and addressing this improves overall model performance. The results also reinforce the importance of key clinical factors such as gestational age, birth weight, and neonatal complications in determining length of stay. Earlier exploratory analysis and clustering further supported these findings by identifying groups of infants with differing levels of prematurity and clinical complexity, which were associated with varying hospitalization durations. Together, these results highlight consistent patterns in how patient characteristics influence NICU outcomes.

These findings have important practical implications. More accurate predictions of NICU length of stay can support hospital resource planning, improve staffing decisions, and assist clinicians in anticipating patient needs. Understanding which factors are most strongly associated with longer hospitalizations may also help guide early interventions for high-risk infants. In future work, additional modeling approaches or variables could be explored to further improve prediction accuracy. For example, incorporating cluster-based groupings of infants, rather than relying solely on individual predictors such as gestational age, may better capture complex clinical interactions and improve predictive performance. Another important consideration is the

inclusion of infants who did not survive in the analysis, as these cases may have either very short or prolonged lengths of stay depending on the timing and severity of illness. This introduces additional variability and can affect model interpretation, suggesting that survival status could be examined separately or incorporated into alternative modeling approaches to better account for these infants.

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Appendices

Appendix A: Data Dictionary

The names and descriptions of the variables were obtained from de-identified neonatal intensive care unit (NICU) data collected at Duke University Hospital. These variables were originally used to support a simulation model developed by Dr. Chris DeRienzo and collaborators at SAS Institute. Below are the 12 variables used in this analysis.

patient_id	De-identified, randomized patient identification number assigned to each infant
survival_status	Whether the infant was alive or deceased at the time of discharge
gestational_age	Gestational age at birth (in weeks)
birth_weight	Birth weight of the infant (in grams)
gender	Sex of the infant
inborn	Whether the infant was born at Duke Hospital or transferred from another facility (Yes, No)
race_ethnicity	Racial/ethnic classification of the infant (White, Black, Other)
length_of_stay	Total length of stay (in days) from admission to discharge or death
ivh	Presence or grade of intraventricular hemorrhage (IVH), bleeding into the internal structures of the brain (None, Grade I, Grade II, Grade III, Grade IV)
rop	Presence or stage of retinopathy of prematurity (ROP), a condition affecting blood vessels in the eye that leads to blindness if not promptly treated (None, Stage I, Stage II, Stage III-IV (No Cryo or Laser Surgery), Stage III-IV (With Cryo or Laser Surgery))
nec	Presence or type of necrotizing enterocolitis (NEC), a severe neonatal intestinal condition (None, Medical, Surgical)
pda	Presence and management of patent ductus arteriosus (PDA), a neonatal cardiovascular anomaly requiring medical or surgical intervention (None, None (Medication Only), Yes (Medically Treated), Yes (Ligated), Yes (Untreated))